

Analytic Weyl Anomaly from Plumbing to the Abel–Jacobi Metric

Machine Note

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Purpose. We want the Weyl-anomaly factor that converts a vacuum partition function defined by CFT sewing in a plumbing frame to the metric induced from the flat Abel–Jacobi torus. The direct Polyakov formula is awkward because a plumbing metric is naturally singular at the seams. Instead, we evaluate one real free-boson partition function in the two frames and take their quotient. The result determines the Weyl factor of every CFT by its central charge.

1 Abel–Jacobi map and canonical metric

Let Σ_g be a compact genus- g Riemann surface. Choose a symplectic basis (A_I, B_I) and normalized holomorphic one-forms

$$\oint_{A_J} \omega_I = \delta_{IJ}, \quad \oint_{B_J} \omega_I = \Omega_{IJ}, \quad Y \equiv \text{Im } \Omega > 0. \quad (1)$$

The Abel–Jacobi map is

$$\mathfrak{A}_{p_0} : \Sigma_g \longrightarrow J(\Sigma_g) = \mathbb{C}^g / (\mathbb{Z}^g + \Omega \mathbb{Z}^g), \quad p \longmapsto \left(\int_{p_0}^p \omega_1, \dots, \int_{p_0}^p \omega_g \right). \quad (2)$$

Pulling back the flat Hermitian form $dZ^T Y^{-1} d\bar{Z}$ gives the Bergman, or Abel–Jacobi, metric. We use its unit-area normalization,

$$\boxed{\mu_{\text{can}} = \frac{i}{2g} \sum_{I,J=1}^g (Y^{-1})_{IJ} \omega_I \wedge \overline{\omega_J}, \quad \int_{\Sigma_g} \mu_{\text{can}} = 1.} \quad (3)$$

The unrescaled restriction has area g . This fixed rescaling changes only a moduli-independent determinant constant at fixed genus.

For one noncompact real boson, after dividing out the constant target-space zero mode, write

$$Z_X[g] = \left(\frac{\det' \Delta_g}{\text{Area}(g)} \right)^{-1/2}. \quad (4)$$

If $g' = e^{2\sigma} g$ is smooth, then

$$\log \frac{Z_X[g']}{Z_X[g]} = \frac{1}{24\pi} \int_{\Sigma_g} (|\nabla \sigma|_g^2 + R_g \sigma) dA_g. \quad (5)$$

For plumbing metrics, the quotient below is the regularized definition of the right-hand side, including its seam contributions.

2 Problem and quotient strategy

For a plumbing channel a , define

$$W_a^{\text{pl/can}} \equiv \frac{Z_X^{\text{pl},a}}{Z_X^{\text{can}}}, \quad \mathcal{A}_a \equiv \log W_a^{\text{pl/can}}. \quad (6)$$

Here the plumbing propagator is the raw q^{L_0} propagator: no $q^{-c/24}$ Casimir factor is hidden in $Z_X^{\text{pl},a}$. Momentum states obey $\langle p|p' \rangle = \delta(p - p')$ and completeness uses dp .

Since the Weyl anomaly is proportional to the ordinary Virasoro central charge c , a vacuum partition function transforms as

$$Z_c^{\text{can}} = Z_c^{\text{pl},a} \left(W_a^{\text{pl/can}} \right)^{-c} = Z_c^{\text{pl},a} e^{-c\mathcal{A}_a}. \quad (7)$$

3 Genus one

For $\Sigma_1 = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, the Abel–Jacobi map is the identity and

$$ds_{\text{can}}^2 = \frac{|dz|^2}{\tau_2}, \quad \tau_2 = \text{Im } \tau. \quad (8)$$

With the standard torus CFT normalization,

$$\det ' \Delta_{\text{can}} = \tau_2 |\eta(\tau)|^4, \quad Z_X^{\text{can},(1)}(\tau) = \frac{1}{\sqrt{\tau_2} |\eta(\tau)|^2}. \quad (9)$$

For $q = e^{2\pi i\tau}$, raw plumbing gives

$$Z_X^{\text{pl},(1)} = \frac{1}{\sqrt{\tau_2}} \prod_{n=1}^{\infty} |1 - q^n|^{-2}. \quad (10)$$

Using $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$,

$$W_1^{\text{pl/can}} = |q|^{1/12}, \quad \mathcal{A}_1 = \frac{1}{12} \log |q|. \quad (11)$$

Thus $Z_c^{\text{can}} = Z_c^{\text{pl}} |q|^{-c/12}$. If the sewing propagator already contains $q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - c/24}$, this genus-one transition is instead unity.

4 The bielliptic conformal frame

The analogue of the four-point pillow starts from a torus

$$E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}) \quad (12)$$

with two insertions at z_1, z_2 . A connected double cover $\pi : \Sigma_2 \rightarrow E_\tau$ branched at these points has genus two by Riemann–Hurwitz. Its deck involution ι has two fixed points, and the quotient $\Sigma_2/\langle \iota \rangle$ is the original torus with two angle- π corners. Thus this is a branched conformal frame, rather than an ordinary one-to-one change of coordinate.

Let $E(z, w)$ be the prime form of E_τ ; in the flat coordinate we use

$$E(z, w) = \frac{\theta_1(z - w \mid \tau)}{\theta_1'(0 \mid \tau)}. \quad (13)$$

Choose $a \in E_\tau$ such that

$$2a = z_1 + z_2 \quad \text{in the group law of } E_\tau. \quad (14)$$

There are four choices, differing by $E_\tau[2]$; they encode the four discrete $\mathbb{Z}_2$ monodromy sectors. If $z_1 + z_2 - 2a = m + n\tau$, define

$$f_a(z) = e^{-2\pi i n z} \frac{E(z, z_1)E(z, z_2)}{E(z, a)^2}. \quad (15)$$

The exponential makes f_a doubly periodic. Its divisor is $z_1 + z_2 - 2a$, so the curve

$$\boxed{\Sigma_{2,a} : \quad \xi^2 = f_a(z)} \quad (16)$$

is branched only at z_1, z_2 ; the double pole at a is unramified.

The invariant and anti-invariant holomorphic one-forms are

$$\omega_+ = \pi^* dz, \quad \omega_- = \pi^* \left(\frac{dz}{\xi} \right), \quad \iota^* \omega_\pm = \pm \omega_\pm. \quad (17)$$

In the torus coordinate, the cut differential is

$$\boxed{\omega_-(z) = \mathcal{N}_a e^{\pi i n z} \frac{\theta_1(z - a \mid \tau)}{\sqrt{\theta_1(z - z_1 \mid \tau) \theta_1(z - z_2 \mid \tau)}} dz.} \quad (18)$$

For $z_1 = 0, z_2 = u$, one may take $a = u/2 + \epsilon$ with $\epsilon \in E_\tau[2]$. Formula (18) is the torus replacement of $dz/\sqrt{z(z-1)(z-w)}$. The theta-function numerator is required by double periodicity; it also supplies the zero divisor of the genus-two one-form.

Regularity is transparent locally. At a branch point, write $z - z_i = t^2$. Then $\xi \sim t$ and

$$\omega_- \sim \frac{dz}{t} = 2 dt. \quad (19)$$

At $z = a$, one has $\xi \sim (z - a)^{-1}$ and hence $\omega_- \sim (z - a)dz$. The two lifts of a therefore give the two zeros required by $\deg K_{\Sigma_2} = 2$.

Choose anti-invariant cycles A_-, B_- and set

$$\widehat{\omega}_- = \frac{\omega_-}{\oint_{A_-} \omega_-}, \quad \tau_P(\tau, z_2 - z_1) = \oint_{B_-} \widehat{\omega}_-. \quad (20)$$

The two eigenperiods are the quotient modulus τ and the Prym modulus τ_P . In a common involution-adapted convention,

$$\Omega = \frac{1}{2} \begin{pmatrix} \tau + \tau_P & \tau - \tau_P \\ \tau - \tau_P & \tau + \tau_P \end{pmatrix}. \quad (21)$$

Factors of two can move between (20) and (21) under a different integral homology convention; invariantly, the two eigenperiods are τ and τ_P .

Now present the genus-two surface as two pairs of pants joined by three plumbing tubes e_1, e_2, e_3 , with parameters q_1, q_2, q_3 . The bielliptic involution exchanges e_1, e_2 and preserves e_3 , so equivariant sewing imposes

$$\boxed{(q_1, q_2, q_3) = (x, x, y)}. \quad (22)$$

The quotient graph is the two-edge necklace graph for a torus two-point function: the orbit $\{e_1, e_2\}$ gives one quotient tube, the invariant edge gives the other, and the two fixed points give the external insertions. Equality of q_1, q_2 constrains the moduli, not the two channel labels; the involution exchanges those labels, and an invariant block is obtained by the corresponding projection.

Equations (20) and (22) give an analytic definition of the desired change of variables:

$$\tau_+(x, y) = \tau, \quad \tau_-(x, y) = \tau_{\mathcal{P}}(\tau, u). \quad (23)$$

Equivalently, the two standard necklace multipliers are analytic functions $p_1(x, y), p_2(x, y)$. They need not equal x, y literally: the quotient of an invariant collar can introduce a square or square root, fixed by the local plumbing coordinates.

Finally, the Abel–Jacobi metric diagonalizes in the eigenbasis as

$$ds_{\text{can}}^2 \propto \frac{|\omega_+|^2}{\text{Im } \tau} + \frac{|\widehat{\omega}_-|^2}{\text{Im } \tau_{\mathcal{P}}}, \quad (24)$$

with the overall factor fixed by (3). Although ω_- changes sign across the cut, $|\omega_-|^2$ is single-valued on the quotient. Near a fixed point,

$$|\omega_-|^2 \sim \frac{|dz|^2}{|z - z_i|}, \quad (25)$$

which is the angle- π corner. Thus the prime-form construction supplies the conformal frame relating the torus two-point geometry to $\Sigma_2/\mathbb{Z}_2$, as well as the explicit canonical metric needed for the Weyl-anomaly quotient.

5 Genus two

Let $Y = \text{Im } \Omega$ and use the raw Igusa product

$$\chi_{10}^{\text{raw}}(\Omega) = \prod_{\delta \text{ even}} \theta[\delta](0|\Omega)^2. \quad (26)$$

Define the Petersson-norm combination

$$F_2(\Omega) = (\det Y)^{5/2} \prod_{\delta \text{ even}} |\theta[\delta](0|\Omega)| = (\det Y)^{5/2} |\chi_{10}^{\text{raw}}(\Omega)|^{1/2}. \quad (27)$$

For the Abel–Jacobi metric, Klein–Kokotov–Korotkin give

$$\det' \Delta_{\text{can}} = C_2 F_2(\Omega)^{1/3}, \quad (28)$$

where C_2 is independent of the moduli. Hence

$$\boxed{Z_X^{\text{can},(2)}(\Omega) = C_2^{-1/2} (\det Y)^{-5/12} |\chi_{10}^{\text{raw}}(\Omega)|^{-1/12}}. \quad (29)$$

Let Γ_a be the rank-two Schottky group obtained from plumbing channel a . If k_γ is the attracting multiplier of a primitive class, set

$$\mathcal{P}_a(\mathbf{q}) = \prod_{[\gamma] \in \mathcal{P}_{\text{prim}}} \prod_{n=1}^{\infty} (1 - k_\gamma(\mathbf{q})^n)^{-1}, \quad (30)$$

with one representative of each inverse-paired primitive class. The exact raw plumbing scalar is

$$Z_X^{\text{pl},a}(\mathbf{q}) = (\det Y_a)^{-1/2} |\mathcal{P}_a(\mathbf{q})|^2. \quad (31)$$

The factor $(\det Y_a)^{-1/2}$ is the Gaussian integral over the two handle momenta; it is distinct from the constant target-space zero mode.

Dividing (31) by (29) gives the analytic Weyl factor

$$W_a^{\text{pl}/\text{can}} = C_2^{1/2} (\det Y_a)^{-1/12} |\chi_{10}^{\text{raw}}(\Omega_a)|^{1/12} |\mathcal{P}_a(\mathbf{q})|^2. \quad (32)$$

Equivalently,

$$\mathcal{A}_a = 2 \operatorname{Re} \log \mathcal{P}_a - \frac{1}{12} \log \det Y_a + \frac{1}{12} \log |\chi_{10}^{\text{raw}}(\Omega_a)| + \frac{1}{2} \log C_2. \quad (33)$$

Equations (32)–(33) are the desired analytic answer. The period map $\Omega_a(\mathbf{q})$ and the multipliers $k_\gamma(\mathbf{q})$ have convergent Schottky expansions throughout the plumbing domain.

Relative channel comparison. If channels a and b represent the same surface in the same symplectic marking, then the canonical determinant and C_2 cancel:

$$\frac{W_a^{\text{pl}/\text{can}}}{W_b^{\text{pl}/\text{can}}} = \frac{Z_X^{\text{pl},a}}{Z_X^{\text{pl},b}}. \quad (34)$$

Period matrices recorded in different $\operatorname{Sp}(4, \mathbb{Z})$ markings must first be transported to a common marking, together with the corresponding automorphy factors.

Type 0B application. One real $N = 1$ superfield has $c = 3/2$, so its Weyl factor is $(W_a^{\text{pl}/\text{can}})^{3/2}$. Nine such superfields and the $b = 1$ super-Liouville theory both have $c = 27/2$; their common Weyl factor therefore cancels in a same-frame quotient.

References

1. C. Klein, A. Kokotov, and D. Korotkin, *Extremal properties of the determinant of the Laplacian in the Bergman metric on the moduli space of genus two Riemann surfaces*, [arXiv:math/0511217](#).
2. G. Mason and M. P. Tuite, *The genus two partition function for free bosonic and lattice vertex operator algebras*, [arXiv:0712.0628](#).